Shubham Gehlot

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Education

M.Tech in Artificial Intelligence **Aug. 2023 – June 2025**
Indian Institute of Science, Bangalore *CGPA: 7.0/10*

B.Tech in Electrical Engineering **Aug. 2019 – May 2023**
Maulana Azad National Institute of Technology, Bhopal *CGPA: 8.80/10*

Work Experience

Data Scientist — Lowe's **[July 2025 – Present]**

- Led the development of an automated demand forecasting pipeline using ONNX-based machine learning models, generating weekly forecasts across promotional and non-promotional periods for downstream analytics and financial planning
- Applied supervised machine learning regression models to estimate baseline demand during non-promotional periods and integrated forecasts across promotion, item, and weekly aggregation levels, improving forecast consistency and reducing reconciliation discrepancies by **8–10%**.
- Conducted root-cause analysis of forecast variances and aggregation mismatches across multiple forecasting hierarchies, identifying data quality and reconciliation issues that improved alignment between ML forecasts and financial calculations by **4.5%**.

Projects

AI Travel Itinerary Planner via Multi-Agent System **[Feb 2026]**

- Designed and developed a multi-agent travel planning system using LangGraph, **orchestrating 7 specialized agents** through a shared workflow with **11 state variables for end-to-end itinerary generation**.
- Integrated **6 Llama 3.2-powered agents via Ollama** for fully local inference, eliminating external LLM API costs while enabling privacy-preserving travel recommendations.
- Built a Streamlit application **supporting 7 travel preference inputs** and generating personalized itineraries for trips of up to 30 days, including weather insights, packing suggestions, activity recommendations, and cultural highlights.

LangGraph-Based Multi-Agent Time Series Prediction System **[Sept 2025]**

- Built a hybrid time-series forecasting system using LangGraph and LLaMA 3.2, orchestrating a 5-node pipeline to process **365+ days of retail sales data and generate 7-day forecasts.**
- Evaluated Prophet and LLM-based forecasting models using RMSE, MAE, and SMAPE, achieving **up to 18% lower prediction error** with statistical forecasting baselines.
- Engineered a modular **StateGraph workflow** for preprocessing, forecasting, evaluation, and explainability, **reducing experimentation time by 60%** and enabling reproducible forecasting pipelines.

3D Reconstruction and View Synthesis Using NeRF **[April 2025]**

- Developed a **Neural Radiance Field (NeRF)** model for 3D reconstruction and view synthesis on the "Fern" and "Flower" datasets, training up to **200,000 iterations**.
- Achieved superior rendering quality with **PSNR of 28.70 dB (Fern) and 30.81 dB (Flower)**, significantly outperforming traditional optical flow methods. Automated generation of novel views through interpolated camera poses, enabling smooth and realistic scene transitions.

Safe Biomedical Natural Language Inference for Clinical Trials **[March 2024]**

- Developed a **Natural Language Inference (NLI) pipeline** for Clinical Trial Reports (CTRs), targeting multi-hop reasoning and numerical inference across long-form biomedical text (2–5 evidence hops per instance).
- Fine-tuned transformer-based models including **DeBERTa-v3 and XLNet**, achieving strong performance on textual entailment with **F1 scores exceeding 0.78** across multiple clinical reasoning categories.
- Benchmarked Large Language Models using zero-shot and Chain-of-Thought (CoT) prompting, and achieved a **best F1 score of 0.807** with DeBERTa on the NLI task, outperforming XLNet and LLM-based baselines by **4–6%**.

Technical Skills

Languages:C/C++, Python, SQL,
Python Libraries: Numpy, Pandas, Pyspark, Matplotlib, Sklearn, Seaborn, Pytorch, XgBoost, LightGBM, Streamlit, FPDF, Google Serper API, python-dotenv
Miscellaneous: Hugging Face, Exploratory Data Analysis, Machine Learning, Deep Learning, Bigquery, GCP, Time-Series Forecasting, Airflow, LangGraph, LangChain, Ollama, Prompt Engineering, Multi-Agent Systems